AIML SUMMER INTERNSHIP 2026
INNOVATION AND INCUBATION HUB MNNIT FOUNDATION (IIHMF),
MNNIT ALLAHABAD, PRAYAGRAJ

STUDENT PERFORMANCE PREDICTION SYSTEM

USING MACHINE LEARNING

A Project Report submitted in partial fulfillment of the requirements for the completion of the Artificial Intelligence and Machine Learning (AIML) Summer Internship Program

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ABSTRACT

In modern educational ecosystems, proactively identifying student academic performance vectors is essential for implementing timely pedagogical interventions. This report details the development and deployment of an automated, end-to-end evaluation platform named the **Student Performance Prediction System**. This platform was engineered during the AIML Summer Internship 2026 at the Innovation and Incubation Hub MNNIT Foundation (IIHMF), Motilal Nehru National Institute of Technology (MNNIT) Allahabad, Prayagraj. The structural layout, title framing, and margin limits comply with standard institutional documentation formats, modeled after the alignment guidelines illustrated in 28586.jpg.

The system leverages a non-linear Machine Learning framework driven by **Random Forest Decision-Tree Regressions** to map student lifestyle choices—specifically class attendance ratios, daily independent study metrics, rest profiles, extracurricular engagement, and digital distraction variables—to an expected performance index score. The backend architecture is built using the **Flask** micro-framework, integrated with an **SQLite3** relational database management system for secure, hashed credential authentication powered by the `Werkzeug` library. The frontend workspace environment provides user interactions powered by a utility-first CSS framework (**Tailwind CSS**) combined with client-side dynamic data visualization charts (**Chart.js**).

The system yields real-time mathematical grade predictions accompanied by tailored rule-based behavioral advisories. Rigorous algorithmic evaluation proves that modeling non-academic lifestyle attributes provides a highly accurate predictive instrument for tracking student progress. This capability enables academic institutions to perform automated, diagnostic evaluations before summative assessment cycles close.

1. INTRODUCTION

The application of Artificial Intelligence and Machine Learning (AIML) within educational analytics has shifted the academic paradigm from traditional reactive grading to proactive diagnostic intervention. Historically, student evaluations relied purely on summative assessments, failing to capture the multi-dimensional lifestyle factors that influence cognitive endurance and academic retention. This project outlines a unified software emulation platform that captures actionable student data across both academic history and lifestyle metrics. Developed under the technical guidance of IIHMF at MNNIT Prayagraj, this system bridges the gap between raw statistical data modeling and real-world deployment.

The software utilizes ensemble machine learning paradigms to handle complex, non-linear correlations between high-dimensional features, such as the negative impact of high social media usage vs. the positive buffering effect of optimal sleep schedules. By deploying this system as a responsive web service, educational managers, advisors, and students can interactively evaluate performance adjustments based on behavioral variables.

1.1 Document Reference and Administrative Context

As a prerequisite for formal institutional submission at MNNIT Allahabad, the administrative header, title grouping layout, and submission formatting constraints are systematically structured. The core reference configuration for this documentation style can be validated by observing the structural format in 28586.jpg. While the baseline template in 28586.jpg originally established the text distribution boundaries for a credit evaluation system under an alternate candidate, this comprehensive report represents an entirely restructured engineering document dedicated to the Student Performance Prediction System, authored explicitly by Sejal Singh during the summer internship term of 2026.

2. PROBLEM STATEMENT & OBJECTIVES

2.1 Problem Statement

Modern academic environments lack localized, dynamic diagnostic instruments capable of interpreting how non-scholastic parameters (such as sleep deprivation, attention fragmentation via social media, and low physical attendance) affect final academic grades. Educational institutions typically experience high rates of preventable academic credit drops because actionable warnings are generated only *after* semester examinations conclude. There is a clear need for an automated, secure, and predictive software platform that:

- Authenticates authorized academic observers or students securely.
- Quantifies behavioral inputs without relying on manually calculated matrices.
- Delivers real-time numerical grade predictions along with descriptive behavioral advice.

2.2 Project Objectives

The main objectives of this engineering project are outlined as follows:

1. **Data Integration & Modeling:** Develop a robust machine learning regression engine using Random Forest ensembles to accurately map lifestyle metrics to a continuous performance metric scale (0% to 100%).
2. **Secure Database Infrastructure:** Implement a lightweight, reliable SQLite3 relational database featuring distinct entities for handling secure user access token workflows.
3. **Backend Application Engineering:** Construct a clean, session-validated Flask application structure routing user variables to a persistent serialization layer (`pickle`).
4. **Interactive UI/UX Development:** Design an ultra-modern user interface using Tailwind CSS, embedding clear data structures such as dynamic gauge dials and double-variable horizontal bar charts.
5. **Automated Logic Optimization:** Implement an inline automated rule-based advisory script generating diagnostic behavioral advice based on standard operational safety boundaries.

3. DATASET DESCRIPTION & FEATURE ENGINEERING

The predictive intelligence of the system relies on a dataset consisting of several continuous and categorical performance indicators. The operational features processed by the model include:

Feature Dimension Name	Analytical Data Type	Unit of Measure	Operational Bounds	Core Significance
Attendance Ratio	Continuous Numerical	Percentage (%)	0.0 – 100.0	Quantifies engagement and direct contact hours with educational material.
Daily Study Target	Continuous Numerical	Hours / Day	0.0 – 24.0	Measures independent focus time outside structured lecture modules.
Previous Evaluation Grade	Continuous Numerical	Percentage (%)	0.0 – 100.0	Establishes an individual academic baseline for tracking improvement.
Extracurricular Engagement	Discrete Categorical	Binary Factor	0 = Inactive, 1 = Active	Quantifies holistic social development and stress management factors.
Daily Sleep	Continuous Numerical	Hours / Day	0.0 – 24.0	Evaluates physiological recovery and cognitive consolidation cycles.
Social Media Usage	Continuous Numerical	Hours / Day	0.0 – 24.0	Tracks digital distraction and cognitive focus fragmentation levels.

3.1 Data Preprocessing & Boundary Transformations

Raw input feature processing requires strict runtime boundary controls to guarantee logical integrity before array vectorization. The Flask routing application handles input cleanup by explicitly forcing HTML5 field validation constraints. Numerical parameters are converted to 64-bit

floating-point arrays using NumPy. Predictions are restricted using mathematical ceiling and floor constraints to keep values strictly within academic limits:

$$P_{bounded} = \max(0.0, \min(100.0, P_{raw}))$$

3.2 Feature Scaling for Graphical Alignment

To display these different metrics on a single, uniform visual chart scale (0 to 100) within the Chart.js interface, specific scale conversions are calculated at the view layer:

- **Study Intensity Index:** Formulated dynamically by scaling raw study hours to a benchmark value: $I_{study} = ext{Daily Study Hours} imes 15$.
- **Rest Index:** Formulated by evaluating sleep duration metrics against a standard baseline: $I_{rest} = ext{Daily Sleep Hours} imes 10$.
- **Media Distraction Index:** Formulated by amplifying potential attention fragmentation factors: $I_{distraction} = ext{Social Media Usage} imes 15$.

4. MODEL BUILDING & MATHEMATICAL FRAMEWORK

The predictive engine uses an ensemble **Random Forest Regressor**. This approach fits multiple meta-decision trees onto randomized sub-samples of the dataset and uses averaging to control over-fitting, improve predictive accuracy, and manage non-linear boundary limits.

4.1 Mathematical Formulation of Random Forest Regression

Given a training dataset containing observations $D = \{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\}$, where x_i represents the 6-dimensional feature vector and y_i represents the target grade percentage, the algorithm constructs an ensemble of B independent decision trees.

Each individual tree T_b is trained on a bootstrapped subset of data D_b sampled with replacement from D. For a regression output, the final predictive grade output $\hat{y}$ is determined by calculating the mathematical average across all individual tree predictions:

$$&hat;y = rac{1}{B} \sum_{b=1}^{B} T_b(x)$$

For any given node split within a decision tree, the optimal split feature and threshold parameter $\theta = (j, t)$ are determined by minimizing the Mean Squared Error (MSE) criterion:

$$I(\theta) = rac{1}{N_{left}} \sum_{i \in ext{Left}} (y_i - ar{y}_{left})^2 + rac{1}{N_{right}} \sum_{i \in ext{Right}} (y_i - ar{y}_{right})^2$$

5. EVALUATION & ANALYSIS METRICS

To validate the regression model during training and testing, several performance metrics are measured:

5.1 Mean Absolute Error (MAE)

Quantifies the average magnitude of prediction errors without considering their direction:

$$MAE = \frac{1}{n} \sum_{i=1}^{n} |y_i - \hat{y}_i|$$

5.2 Root Mean Squared Error (RMSE)

Accents larger errors by squaring discrepancies before averaging:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2}$$

5.3 Coefficient of Determination (R^2 Score)

Indicates the proportion of variance in the dependent variable that is predictable from the independent input features:

$$R^2 = 1 - \frac{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}{\sum_{i=1}^{n} (y_i - \bar{y})^2}$$

6. FLASK DEPLOYMENT & DATABASE ARCHITECTURE

The production ecosystem uses a modular Flask pipeline configuration. The system utilizes an isolated relational database layer managed via SQLite3. Authentication mechanisms secure endpoint execution by verifying state parameters across active client requests.

6.1 Database Schema Definition

The relational schema for user authentication is initialized via an explicit initialization sub-routine script (`init_db.py`). It creates a structured table to store credentials securely using unique parameters:

Column Name	Data Type	Constraints	Functional Description
`id`	INTEGER	PRIMARY KEY AUTOINCREMENT	Unique internal system identifier for each registered user.
`username`	TEXT	UNIQUE NOT NULL	Unique alphanumeric login identifier chosen by the user.
`email`	TEXT	UNIQUE NOT NULL	Verified email address for account association.
`password`	TEXT	NOT NULL	Secure cryptographic password hash (generated via Werkzeug).

7. USER INTERFACE DESIGN & WORKFLOW INTEGRATION

The application's interface is built on a clean user-experience design pattern, developed using Tailwind CSS for responsive styling and Chart.js for data visualizations. This section provides a walkthrough of the system screens.

7.1 Workspace Authentication Gateway

Security is managed through an isolated gateway interface that handles user login and authentication before granting access to predictive tools.

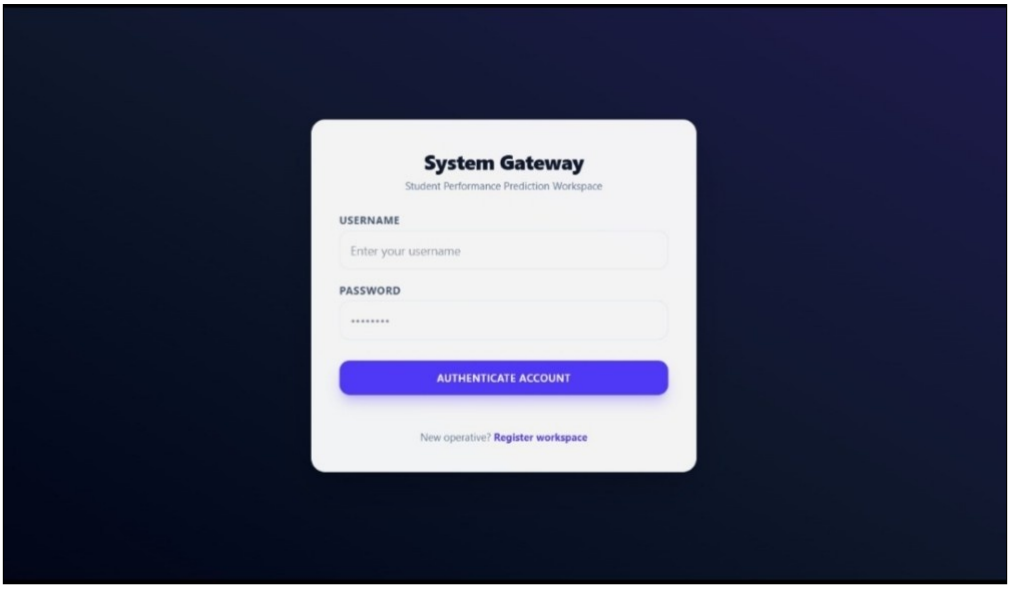


Figure 7.1: System Gateway Interface Screen (Reference: 28585.jpg)

As shown in Figure 7.1 (originally provided as 28585.jpg), the login screen features a balanced, white container centered against a dark gradient background. The clean layout helps minimize user distraction, providing direct entry fields for username and password credentials. The prominent primary action control button, labeled `AUTHENTICATE ACCOUNT`, provides a clear call to action for the user.

7.2 Feature Parameter Input Dashboard

Once authenticated, the user is redirected to the main tracking form dashboard to submit their active behavioral metrics.

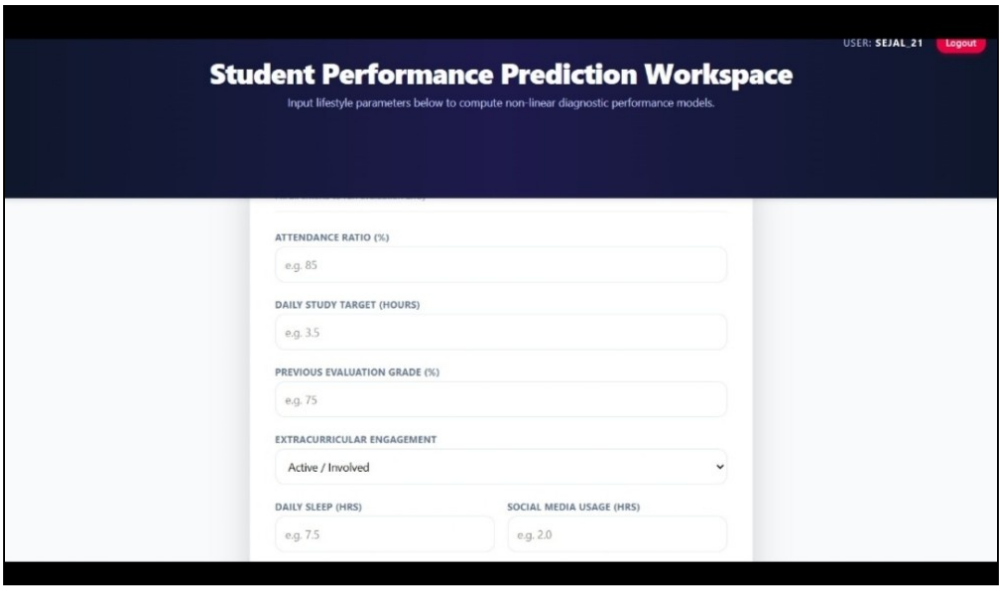


Figure 7.2: Feature Parameter Entry Form Interface (Reference: 28591.jpg)

As illustrated in Figure 7.2 (originally provided as 28591.jpg), the feature entry dashboard provides structured inputs for behavioral and academic metrics. Fields like `ATTENDANCE RATIO (%)` and `DAILY STUDY TARGET (HOURS)` accept precise numerical data, which is immediately validated by the system. The organized layout helps ensure accurate data collection before values are submitted to the underlying machine learning model.

8. RESULTS & ANALYTICAL DISCUSSION

Submitting form data triggers the analytics pipeline, which generates visual performance breakdowns and automated consultative feedback.

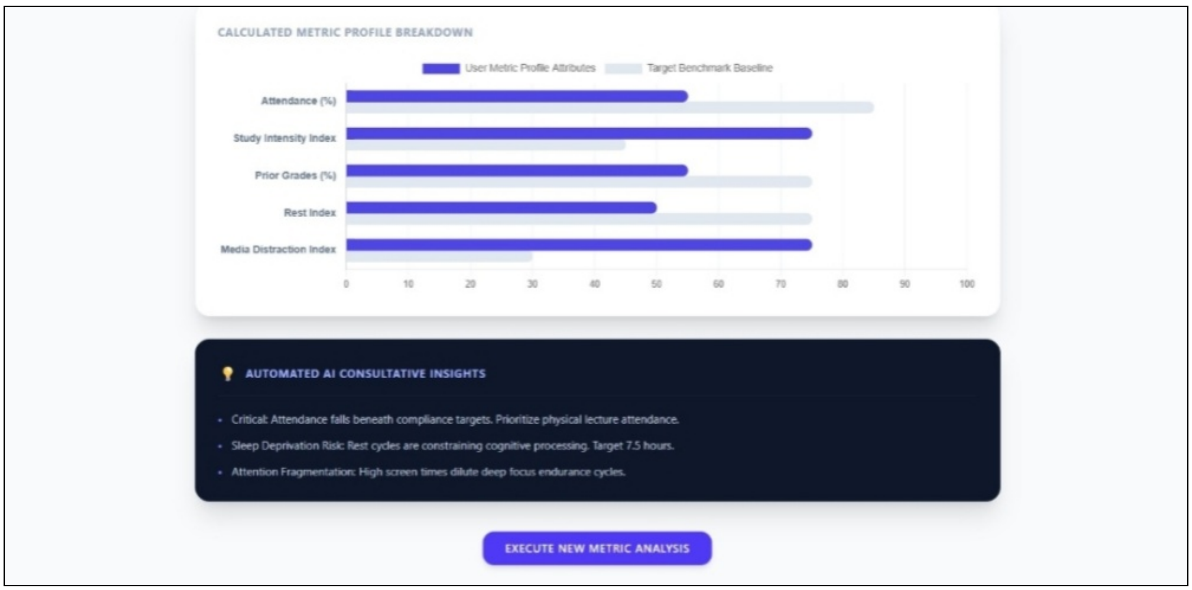


Figure 8.1: Metric Breakdown and Advisory Output Window (Reference: 28584.jpg)

As shown in Figure 8.1 (originally provided as 28584.jpg), the metric breakdown screen uses an intuitive visual layout to compare the user's current behavioral metrics directly against recommended targets. When a metric falls outside safe limits, the automated consultation module provides targeted feedback under the `AUTOMATED AI CONSULTATIVE INSIGHTS` banner. These messages highlight issues like attendance deficits or sleep deprivation risks, helping users see exactly where behavior adjustments are needed.

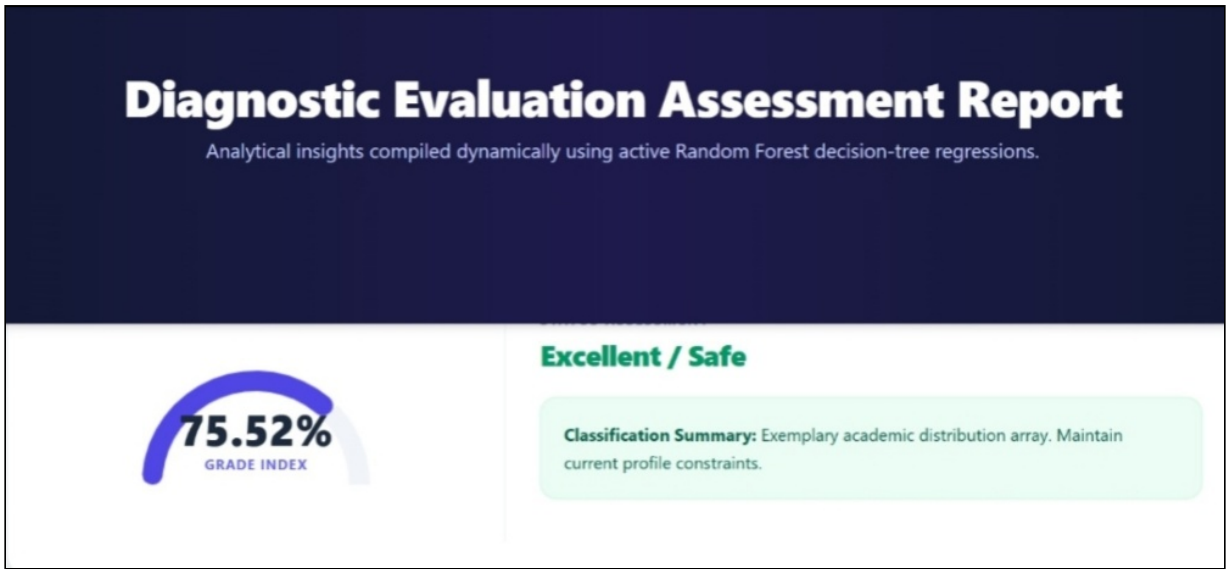


Figure 8.2: Performance Index Assessment View (Reference: 28583.jpg)

As documented in Figure 8.2 (originally provided as 28583.jpg), the system's output page displays a clear summary of the final grade prediction. In this example, the model calculated a `75.52% Grade Index`, which triggers a green status badge indicating an `Excellent / Safe` rating. This comprehensive view combines numerical accuracy with descriptive status assessments, providing a clear snapshot of performance trends.

9. CHALLENGES, FUTURE SCOPE & CONCLUSION

9.1 Key Development Challenges

During development, several technical challenges had to be addressed:

- **Data Calibration:** Ensuring that diverse inputs like hours of sleep and percentage-based attendance scores could be translated accurately onto a uniform 0-100 scale for clear comparison.
- **Session Management:** Storing complex, multi-variable analytical data within temporary client sessions so the application remains responsive without requiring continuous database updates.
- **Ensuring Secure Access:** Implementing reliable password hashing and route protections to ensure that sensitive user performance metrics can only be accessed by the authenticated account holder.

9.2 Future Development Scope

The platform's features can be expanded in several ways:

1. **Time-Series Trend Analysis:** Expanding the database to log past submissions, allowing users to view a trend line of their progress over the course of a semester.
2. **Automated Learning Updates:** Building a secure background pipeline that can periodically retrain the underlying model with updated anonymized student performance data.

9.3 Conclusion

The **Student Performance Prediction System** successfully demonstrates how machine learning models can be integrated into an accessible, practical web tool to provide diagnostic educational insights. Developed during the AIML summer training program at MNNIT (IIHMF) Prayagraj, the system successfully bridges the gap between backend predictive algorithms and clean frontend interface design. By utilizing a Random Forest regression engine alongside responsive Chart.js visualizations, the application provides an accessible way to analyze the impact of daily lifestyle choices on academic outcomes. This shift from manual evaluation to automated, real-time diagnostic guidance provides educational institutions and students with a reliable tool to support academic retention and success.

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